



Solutions:

1. A. $y = \csc x(x \ln x - x + C)$

Using Calculator: Casio FX570ES+

Assign: $x = 0.5$, $y = 1$, $y' \rightarrow A$

Radian mode

$$\sin xy' + \cos xy = \ln x$$

$$\sin(0.5)A + \cos(0.5) = \ln(0.5)$$

$$A = y' = -3.276$$

Solve for C in the choices:

For $y = \csc x(x \ln x - x + C)$:

$$y = \csc x(x \ln x - x + C)$$

$$1 = \frac{1}{\sin(0.5)}(0.5 \ln(0.5) - 0.5 + C)$$

$$C = 1.326$$

Then solve for y' if it match with the given:

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{1}{\sin(x)} (x \ln(x) - x + 1.326) \right) \Bigg|_{x=0.5}$$

$$\frac{dy}{dx} = -3.276 \text{ (it matches)}$$

2. D. e^x

$$(y^2 - x)dx + 2ydy = 0$$

Divide both sides by $2ydx$:

$$\frac{y}{2} - \frac{x}{2y} + \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} + \frac{1}{2}y = \frac{x}{2}y^{-1}$$

$$P(x) = \frac{1}{2}$$

$$Q(x) = \frac{x}{2}$$

$$n = -1$$

$$I.F. = e^{(1-n) \int P dx}$$

$$I.F. = e^{(1+1) \int \frac{1}{2} dx}$$

$$I.F. = e^x$$

3. A. 9

Let x = population at time t

$$\frac{dx}{dt} = \alpha x$$

$$\frac{dx}{x} = \alpha dt$$

Integrate both sides

$$\int \frac{dx}{x} = \int \alpha dt + C$$

$$\ln x = \alpha t + C$$

$$x = Ce^{\alpha t}$$

At $t = 0$, $x = 1$ (initial population)

$$1 = Ce^{\alpha(0)}$$

$$C = 1$$

At $t = 2$, $x = 3$

$$3 = (1)e^{\alpha(2)}$$

$$\alpha = \frac{1}{2} \ln 3$$

At $t = 4$, $x = k$

$$k = e^{\left(\frac{1}{2} \ln 3\right)(4)}$$

$$k = 9$$

Calculator Shortcut:

Mode $\rightarrow$ STAT $\rightarrow e^x$ (5)

$$\begin{array}{|c|c|} \hline x & y \\ \hline 0 & 1 \\ \hline 2 & 3 \\ \hline \end{array} \rightarrow AC$$

4 $\rightarrow$ Shift $\rightarrow$ STAT(1) $\rightarrow$

Reg(5) $\rightarrow \hat{y}(5) \rightarrow = 9$

4. A. $y = \tan\left(\frac{x^2}{2} + c\right)$

$$y' - xy^2 = x$$

$$\frac{dy}{dx} = x(1 + y^2)$$

$$\frac{dy}{1 + y^2} = x dx$$

Integrating both sides:

$$\int \frac{dy}{1 + y^2} = \int x dx + C$$

$$\arctan y = \frac{1}{2}x^2 + C$$

$$y = \tan\left(\frac{1}{2}x^2 + C\right)$$

5. B. $xy = C$

$$x^2 - y^2 = c^2$$

$$2x - 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{x}{y}$$

For orthogonal trajectories:

$$\frac{dy}{dx} = -\frac{y}{x}$$

$$\int \frac{dy}{y} = \int -\frac{dx}{x}$$

$$\ln y = -\ln x + C$$

$$\ln y + \ln x = C$$

$$\ln yx = C$$

$$xy = C$$

6. A. $y = \frac{1}{2}e^{-2t} + \frac{1}{2}$ *

$$\frac{dy}{dt} + 2y = 1$$

$$ye^{\int P dt} = \int Qe^{\int P dt} dt + c$$

$$ye^{\int 2 dt} = \int (1)e^{\int 2 dt} dt + c$$

$$ye^{2t} = \frac{1}{2}e^{2t} + c$$

$$\text{at } t = 0, y = 1$$

$$c = \frac{1}{2}$$

$$ye^{2t} = \frac{1}{2}e^{2t} + \frac{1}{2}$$

$$y = \frac{1}{2} + \frac{1}{2}e^{-2t}$$

7. A. $y_p = -\frac{3}{20} \sin(2x) + \frac{1}{20} \cos(2x)$

$$y_p = A \cos 2x + B \sin 2x$$

$$y_p' = -2A \sin 2x + 2B \cos 2x$$

$$y_p'' = -4A \cos 2x - 4B \sin 2x$$

$$y'' - y' - 2y = \sin 2x$$

$$-4A \cos 2x - 4B \sin 2x + 2A \sin 2x$$

$$-2B \cos 2x$$

$$-2(A \cos 2x + B \sin 2x) = \sin 2x$$

$$-4A - 2B - 2A = 0$$

$$-6A - 2B = 0$$



$$-4B + 2A - 2B = 0$$

$$2A - 6B = 1$$

$$A = \frac{1}{20}$$

$$B = -\frac{3}{20}$$

$$y_p = \frac{1}{20} \cos 2x - \frac{3}{20} \sin 2x$$

8. D. Parabolic

9. B. parabolic

10. D. $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$

11. C. 1

$$x^2z - 2yz^2 + xy = 0$$

$$2xz \frac{\partial x}{\partial z} + x^2 - 4yz + y \frac{\partial x}{\partial z} = 0$$

At (1,1,1)

$$\frac{\partial x}{\partial z} = 1$$

12. A. 15 min

$$\frac{dy}{dt} = \text{rate in} - \text{rate out}$$

$$\frac{dy}{dt} = \left(3 \frac{g}{L}\right) \left(4 \frac{L}{\text{min}}\right) - \left(4 \frac{L}{\text{min}}\right) \left(\frac{y}{40 L}\right)$$

$$\frac{dy}{dt} = 12 - \frac{y}{10}$$

$$10 \frac{dy}{dt} = 120 - y$$

$$\frac{10dy}{120 - y} = dt$$

Integrate both sides:

$$-10 \ln(120 - y) = t + C$$

at $t = 0$, $y = 2 \frac{g}{L} (40 L)$

$$C = -36.89$$

$$-10 \ln(120 - y) = t - 36.89$$

at $y = 111.1$

$$t = 15 \text{ min}$$

13. A. Fourth order, first degree

Fourth order, since the highest derivative in the equation is 4th derivative. First degree, since the exponent or power of the 4th derivative is 1.

14. B. $xy = \frac{x^4}{4} + C$

$$\frac{dy}{dx} + \left(\frac{1}{x}\right)y = x^2$$

Note: This equation is a linear DE:

$$y \text{ (i.f.)} = \int Q(x) \text{ (i.f.)} dx + C \rightarrow \text{Gen. Eq.}$$

where: $P(x) = \frac{1}{x}$; $Q(x) = x^2$

$$\text{i.f.} = e^{\int P(x) dx}$$

$$\text{i.f.} = e^{\int \frac{dx}{x}}$$

$$\text{i.f.} = e^{\ln x}$$

$$\text{i.f.} = x$$

Substitute:

$$y(x) = \int x^2(x) dx + C$$

$$xy = \frac{x^4}{4} + C$$

15. C. $2y dx = (x^2 + 1) dy$

A DE is a variable separable if the coefficient of dx shall be functions of x only and the coefficient of dy shall be functions of y only.

From the choices, $2y dx = (x^2 + 1) dy$ is a variable separable DE

$$2y dx = (x^2 + 1) dy$$

$$\frac{2y dx}{y(x^2 + 1)} = \frac{(x^2 + 1) dy}{y(x^2 + 1)}$$

$$\left(\frac{2}{x^2 + 1}\right) dx = \left(\frac{1}{y}\right) dy$$

16. B. 116 years

Let: P = number of population

$$\frac{dP}{dt} = kP$$

$$\int \frac{dP}{P} = \int k dt$$

$$\ln P = kt + C \rightarrow \text{Gen. Eq.}$$

when $t = 0$; $P = P_0$

$$\ln P_0 = k(0) + C$$

$$C = \ln P_0$$

when $t = 50$ years; $P = 2P_0$

$$\ln 2P_0 = k(50) + \ln P_0$$

$$k = 0.0138629$$

when $t = ?$ $Q = 5P_0$

$$\ln 5P_0 = 0.0138629(t) + \ln P_0$$

$$t = 116 \text{ years}$$

17. A. The order of the highest derivative that appears in the equation.

18. C. $\frac{1}{x^2}$ *

An integrating factor for an ordinary differential equation of the form $Mdx + ndy = 0$ is a function of the form $J(x,y)$ such that $JMdx + JNdy = 0$ is exact that is,

$$\frac{\partial(JM)}{\partial y} = \frac{\partial(JN)}{\partial x}$$

The given equation is not exact since

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = \frac{-2}{x} - 2xy \cos y$$

$$= \frac{-2}{x} (1 + x^2 y \cos y) \neq 0$$

But since

$$r(x) = \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{-2}{x} \text{ is a}$$

function of x , an integrating factor is given by

$$J(x) = e^{-\int r(x) dx}$$

$$= e^{-\int \frac{2}{x} dx}$$

$$= e^{-2 \ln x}$$

$$= \frac{1}{x^2}$$

19. A. $C_1 e^{-2x} + C_2 e^{-3x} + \frac{e^x}{12}$



$$(D^2 + 5D + 6)y = e^x$$

$$m^2 + 5m + 6 = 0$$

$$m = -2, -3$$

$$C.F. = C_1 e^{-2x} + C_2 e^{-3x}$$

$$P.I. = \frac{1}{D^2 + 5D + 6} e^x$$

$$= \frac{1}{12} e^x$$

$$y = C.F. + P.I.$$

$$y = C_1 e^{-2x} + C_2 e^{-3x} + \frac{1}{12} e^x$$

20. D. $v = 50 - 2e^{-0.196t}$ *

To find the solution to an IVP, first find the general solution of the DE and then use the initial condition to identify the exact solution. We get

$$v = 50 + ce^{-0.196t}$$

$$\text{For } v(0) = 48$$

$$48 = v(0) = 50 + c$$

$$c = -2$$

Therefore, the actual solution to the IVP is $v = 50 - 2e^{-0.196t}$

21. B. $C_1 e^{2x} + C_2 e^{3x} + \frac{e^{4x}}{2}$ *

$$(D^2 - 5D + 6)y = e^{4x}$$

$$m^2 - 5m + 6 = 0$$

$$(m - 3)(m - 2) = 0$$

$$m = 2, 3$$

$$C.F. = C_1 e^{2x} + C_2 e^{3x}$$

$$P.I. = \frac{1}{D^2 - 5D + 6} e^{4x}$$

$$= \frac{1}{16 - 20 + 6} e^{4x}$$

$$= \frac{1}{2} e^{4x}$$

$$y = C.F. + P.I.$$

$$y = C_1 e^{2x} + C_2 e^{3x} + \frac{1}{2} e^{4x}$$

22. C. $y = \cos x + 2 \sin x$

$$y'' + y = 0$$

$$(D^2 + 1)y = 0$$

$$m = \pm i$$

$$c.f. = (C_1 \cos x + C_2 \sin x)$$

$$\text{Putting } x = 0, y(0) = 1 \rightarrow C_1 = 0$$

$$\text{Putting } x = \frac{\pi}{2}, y\left(\frac{\pi}{2}\right) = 2 \rightarrow C_2 = 2$$

$$\text{Therefore: } y = \cos x + 2 \sin x$$

23. A. degree 1 order 2

24. D. Ricatti equation

25. C. Bernoulli equation